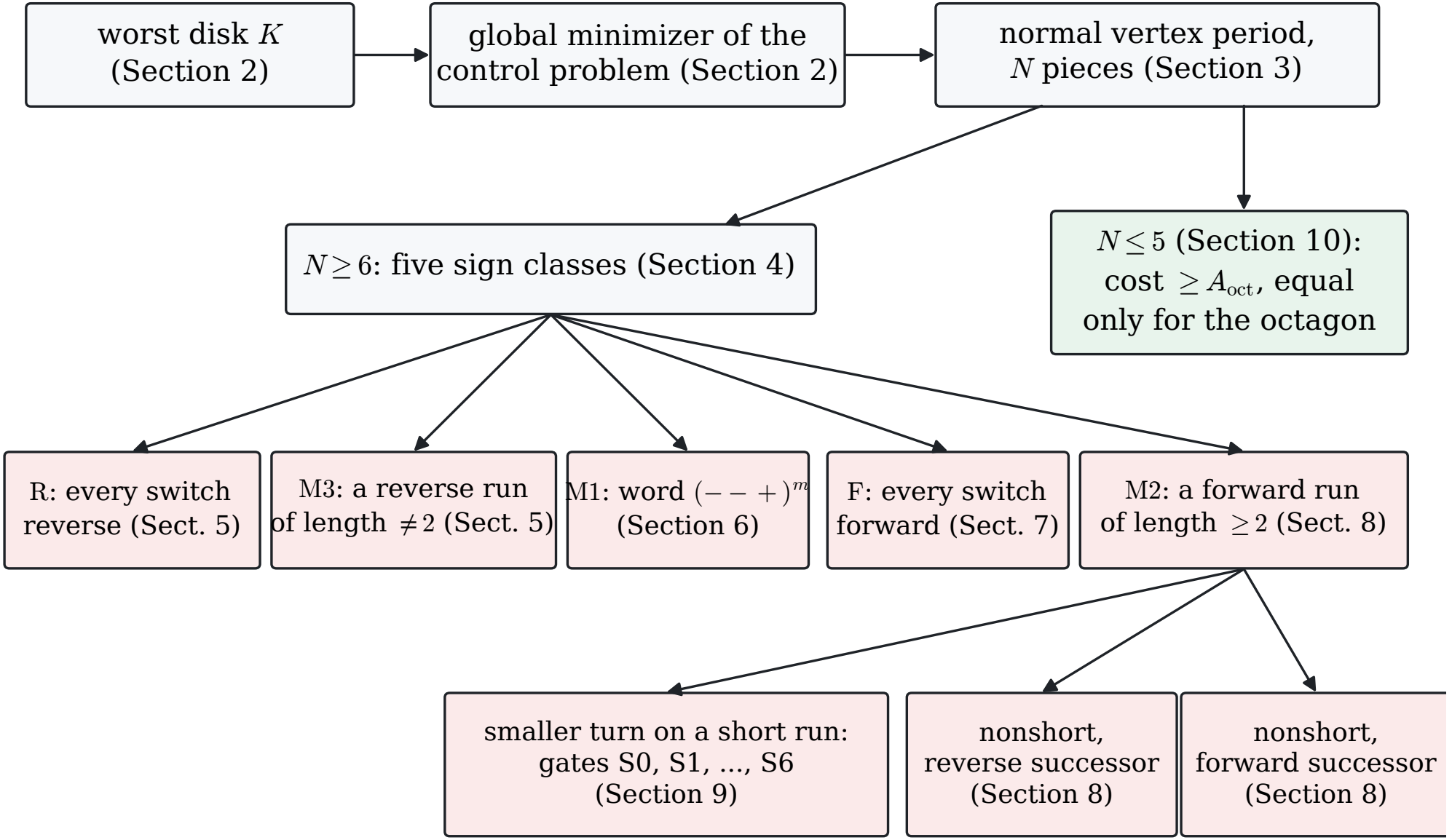
worst disk K (Section 2)
global minimizer of the control problem (Section 2)
normal vertex period, N pieces (Section 3)
$N \geq 6$: five sign classes (Section 4)
$N \leq 5$ (Section 10): cost $\geq A_{\mathrm{oct}}$, equal only for the octagon
R: every switch reverse (Sect. 5)
M3: a reverse run of length $\neq 2$ (Sect. 5)
M1: word $(--+)^m$ (Section 6)
F: every switch forward (Sect. 7)
M2: a forward run of length ≥ 2 (Sect. 8)
smaller turn on a short run: gates S0, S1, ..., S6 (Section 9)
nonshort, reverse successor (Section 8)
nonshort, forward successor (Section 8)